

Project Design Phase Solution Architecture

Date	07/07/2026
Team ID	SWUID2026196344
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between user requirements and the technical implementation of the Nutrition Assistant system. It defines how different software components interact to provide a secure, scalable, and user-friendly nutrition management platform.

The **Nutrition Assistant** is built using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**. The application follows a client-server architecture where users interact with the React frontend, which communicates with the Express.js backend through RESTful APIs. The backend processes requests, authenticates users using JWT, and stores data securely in MongoDB.

The system supports three user roles:

- **User** – Manages personal profile, views meal plans, tracks nutrition and progress.
- **Dietitian** – Creates and manages client meal plans and monitors progress.
- **Administrator** – Manages users, dietitians, and overall system operations.

Example - Solution Architecture Diagram:

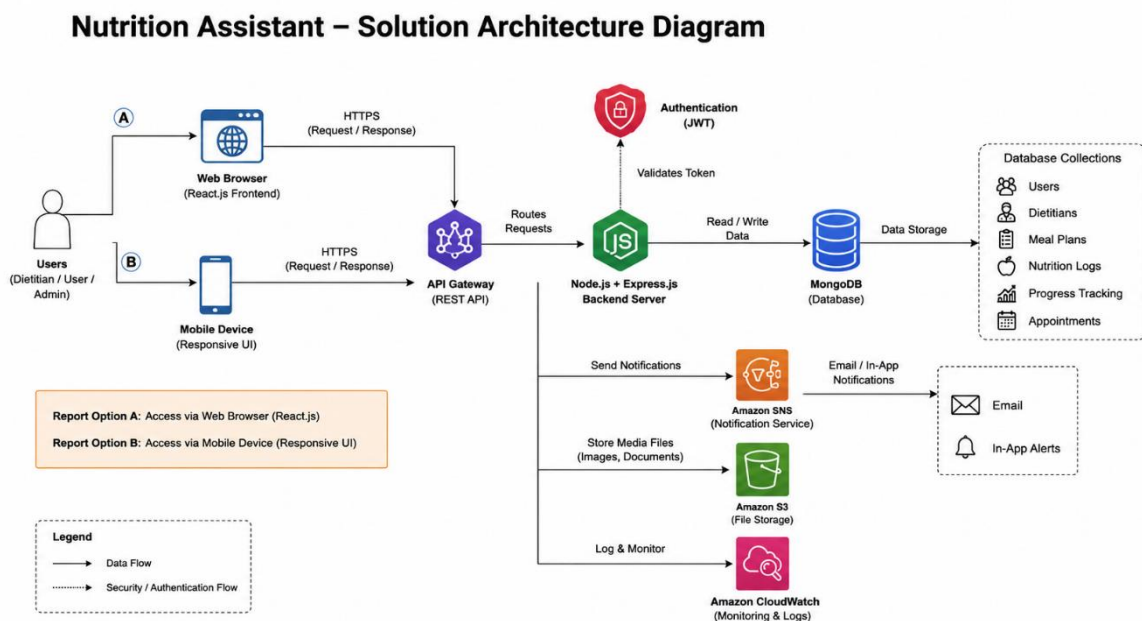


Figure 1: Solution Architecture and Data Flow of the Nutrition Assistant Application.

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>